

Lecture 7 Second Order Homogeneous ODEs



Lecture 7
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Engineering Mathematics III

ENGE702

Lecture 7: Second Order Homogeneous ODEs

Second order ODEs

In this lecture we will look at methods of solution for a certain class of second order ODEs. Specifically, we will look at **second order linear homogeneous ODEs with constant coefficients**. These are ODEs of the form

$$y'' + ay' + by = 0$$

where a and b are constants.

The form above is 'standard form', where the coefficient in front of the y'' is 1. If we have a second order linear homogeneous ODEs with constant coefficients that is not in standard form, we can always put it in standard form by dividing through by whatever is in front of the y'' (this is never 0, otherwise it would not be second order).

Second order ODEs

The method to solve these is quite simple, and is based on our observation from an earlier lecture that the solution equation

$$y' + ky = 0 \quad \text{is} \quad y = e^{-kx}.$$

Since this essentially is the first order version of our second order equation, it may be reasonable to 'assume' this will be a solution for our second order equation as well, and see what happens. Let's assume that the solution of

$$y'' + ay' + by = 0$$

is

$$y = e^{\lambda x}$$

for some constant λ . Then

$$y' = \lambda e^{\lambda x} \quad \text{and} \quad y'' = \lambda^2 e^{\lambda x}.$$

Second order ODEs

Putting these values into our original equation $y'' + ay' + by = 0$, we have

$$(\lambda^2 + a\lambda + b)e^{\lambda x} = 0.$$

So if we can find a solution to this equation (ie find an appropriate λ to make it work), we have found a solution to our second order differential equation.

Since $e^{\lambda x}$ is never 0, it must mean that $\lambda^2 + a\lambda + b = 0$. This is simply solving a quadratic polynomial!

This equation $\lambda^2 + a\lambda + b = 0$ is called the **characteristic equation** of its corresponding differential equation.

We can do this with the quadratic formula (if necessary), to get solutions

$$\lambda_1 = \frac{1}{2}(-a + \sqrt{a^2 - 4b}) \quad \text{and} \quad \lambda_2 = \frac{1}{2}(-a - \sqrt{a^2 - 4b})$$

giving solutions

$$y_1 = e^{\lambda_1 x} \quad \text{and} \quad y_2 = e^{\lambda_2 x}$$

Second order ODEs

linear combination of dependent variables
first order linear ODE

$$y' + p(x)y = r(x)$$

highest derivative y'

second order linear ODE

$$y'' + p(x)y' + q(x)y = r(x)$$

highest derivative y'' linear combination of dependent variables

when $r(x) = 0$, it is homogeneous ODE
when $r(x) \neq 0$, it is inhomogeneous ODE

First look at homogeneous 2nd order linear ODE with constants as

$$y'' + ay' + by = 0$$

If we assume the solution to the ODE has the form $y = e^{\lambda x}$

Then we can deduce that $y' = \lambda e^{\lambda x}$ and $y'' = \lambda^2 e^{\lambda x}$. Substitute into the ODE to get

$$\lambda^2 e^{\lambda x} + a\lambda e^{\lambda x} + be^{\lambda x} = 0$$

$$\Rightarrow \lambda^2 + a\lambda + b = 0 \quad \text{called the characteristic equation}$$

→ solve the equation for λ_1, λ_2

then get the solutions $y_1 = e^{\lambda_1 x}, y_2 = e^{\lambda_2 x}$

Second order ODEs

Remember that we need two linearly independent solutions for a general solution. We have two solutions above, but these will only give non-proportional real valued solutions if $a^2 - 4b > 0$. In other cases we might have to do some more work. We will show how to get solutions for all possible cases.

Second order ODEs

Case 1: Two real roots (ie $a^2 - 4b > 0$):

In this case the solutions

$$y_1 = e^{\lambda_1 x} \quad \text{and} \quad y_2 = e^{\lambda_2 x}$$

are linearly independent, so we have the general solution:

$$y = c_1 e^{\lambda_1 x} + c_2 e^{\lambda_2 x}$$

Second order ODEs

Example: Find the general solution to the differential equation

$$y'' + y' - 2y = 0$$

Second order ODEs

Example: Find the general solution to the differential equation

$$y'' + y' - 2y = 0$$

This is a second order linear homogeneous equation with constant coefficients. We can find the characteristic equation

$$\lambda^2 + \lambda - 2 = 0.$$

Solving this (either by factorising or using the quadratic formula if necessary) gives $\lambda_1 = 1$ and $\lambda_2 = -2$. The general solution will then be:

$$y = c_1 e^x + c_2 e^{-2x}$$

Case 1, the characteristic equation has two real roots λ_1 and λ_2 .

The solution to the ODE is

$$y(x) = c_1 e^{\lambda_1 x} + c_2 e^{\lambda_2 x}.$$

Example: $y'' + y' - 2y = 0$.

Assume $y = e^{\lambda x}$, then $y' = \lambda e^{\lambda x}$, $y'' = \lambda^2 e^{\lambda x}$

$$\text{so } \lambda^2 e^{\lambda x} + \lambda e^{\lambda x} - 2e^{\lambda x} = 0$$

Characteristic equation is $\lambda^2 + \lambda - 2 = 0$ $a=1, b=1, c=-2$

Use the quadratic formula: $\lambda = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

$$\lambda = \frac{-1 \pm \sqrt{1^2 - 4(1)(-2)}}{2} = \frac{-1 \pm 3}{2}$$

$$\Rightarrow \lambda_1 = 1 \quad \text{and} \quad \lambda_2 = -2$$

So the solution to the ODE $y'' + y' - 2y = 0$ is

$$y(x) = c_1 e^{\lambda_1 x} + c_2 e^{\lambda_2 x} = c_1 e^x + c_2 e^{-2x}.$$

Case 2: If the characteristic equation

$$\lambda^2 + a\lambda + b = 0 \quad \text{has one real root } \lambda,$$

then the general solution is

$$y(x) = c_1 e^{\lambda x} + c_2 x e^{\lambda x}$$

Second order ODEs

Case 2: One real root (ie $a^2 - 4b = 0$):

In this case there is only one value for λ (even though it 'occurs twice'). We have one solution $y = e^{\lambda x}$, but need another. In this case it is also true that $y = xe^{\lambda x}$ is a solution (check!).

Our two solutions are linearly independent (they are not proportional). So a general solution is

$$y = c_1 e^{\lambda x} + c_2 x e^{\lambda x}$$

Second order ODEs

Exercise: Find the general solution to the differential equation

$$y'' + 2y' + y = 0$$

Solve $y'' + 2y' + y = 0$

The characteristic equation is $\lambda^2 + 2\lambda + 1 = 0$

The quadratic formula: $\lambda = \frac{-2 \pm \sqrt{2^2 - 4 \times 1}}{2} = \frac{-2 \pm 0}{2} = -1$, So the general solution to the ODE $y'' + 2y' + y = 0$ is $y(x) = c_1 e^{-x} + c_2 x e^{-x}$.

Second order ODEs

Case 3: Complex conjugate roots (ie $a^2 - 4b < 0$):

In this case there are no real roots to the characteristic equation, but we will get two complex roots. After some manipulation (see the last part of chapter 2.2 in the text book for details), we can obtain linearly independent solutions

$$y_1 = e^{-\frac{a}{2}x} \cos(\omega x) \quad \text{and} \quad y_2 = e^{-\frac{a}{2}x} \sin(\omega x)$$

where

$$\omega^2 = b - \frac{1}{4}a^2.$$

Our general solution then becomes:

$$y = e^{-\frac{a}{2}x} (A \cos(\omega x) + B \sin(\omega x))$$

Note that the complex solutions of the characteristic equation are given by $\lambda = -\frac{a}{2} \pm i\omega$

Case 3: The characteristic equation has two complex roots $\lambda = -\frac{a}{2} \pm i\omega$.

then the solution to the ODE is

$$y = e^{-\frac{a}{2}x} (A \cos(\omega x) + B \sin(\omega x))$$

Second order ODEs

The cases can be summarised as follows: To solve the differential equation

$$y'' + ay' + by = 0,$$

find roots of the characteristic equation

$$\lambda^2 + a\lambda + b = 0.$$

The solutions will be as follows, and depend on the types of roots found.

Case	Roots	General Solution
1	Distinct real λ_1, λ_2	$y = c_1 e^{\lambda_1 x} + c_2 e^{\lambda_2 x}$
2	One real λ	$y = (c_1 + c_2 x) e^{\lambda x}$
3	Complex conjugate $\lambda_1 = -\frac{1}{2}a + i\omega$ $\lambda_2 = -\frac{1}{2}a - i\omega$	$y = e^{-\frac{a}{2}x} (A \cos(\omega x) + B \sin(\omega x))$

Second order ODEs

Example: Find the general solution to the differential equation

$$y'' + 2y' + 5y = 0$$

$$y'' + 2y' + 5y = 0, \quad \sqrt{-1} = i$$

The characteristic equation is $\lambda^2 + 2\lambda + 5 = 0$

using the quadratic formula:

$$\lambda = \frac{-2 \pm \sqrt{2^2 - 4 \times 5}}{2} = \frac{-2 \pm \sqrt{-16}}{2}$$

$$y'' + 2y' + 5y = 0$$

$$\lambda = \frac{-2 \pm \sqrt{2^2 - 4 \times 5}}{2} = \frac{-2 \pm \sqrt{-16}}{2}$$

$$= \frac{-2 \pm 4\sqrt{-1}}{2} = \frac{-2 \pm 4i}{2} = -1 \pm 2i$$

So the general solution is complex roots

$$y(x) = e^{-1x} (A \cos(2x) + B \sin(2x))$$

Second order ODEs

Example: Find the general solution to the differential equation

$$y'' + 2y' + 5y = 0$$

In this case the characteristic equation has roots $-1 \pm 2i$. So $\omega = 2$, and our solution is

$$y = e^{-x} (A \cos(2x) + B \sin(2x))$$

Second order ODEs

Exercises: Find the general solutions to the following differential equations:

► $y'' + 4y' + 3y = 0$

► $y'' + 4y' + 4y = 0$

► $y'' + 4y' + 5y = 0$

Text Book References

Chapter 2.2